Each question is equally weighted. Each question has one correct answer; there are no multi-select questions. There are no penalties for incorrect responses.

Q1 [Git]
In a git repository, Bernardo and Dylan have made conflicting changes to a tracked file. Bernardo pushes his changes first, and these are accepted. When Dylan goes to push, discovering the conflict, he uses "git fetch" and "git merge" to merge these changes, then pushes. Dylan uses "git log --graph –oneline" to see what has happened, which gives an output including.

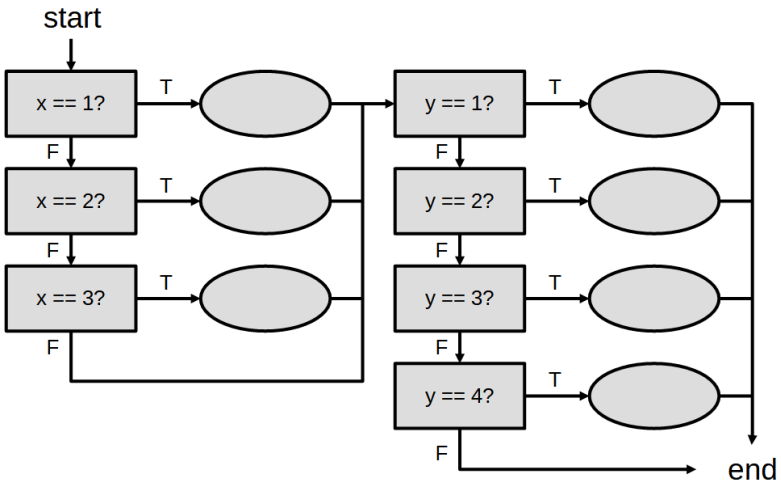
```
*   A (HEAD -> my_branch)
|\
| * B (origin/my_branch)
* | C
|/
*   D
```

Which of these statements is plausible?
a. Commit D was the most recent of the four commits.
b. Commits A and B were created by the "git merge" command.
c. Commit C was pointed to by a remote-tracking branch prior to merging.
d. Commits B and D were made by Bernardo.

Q2 [Design Patterns]
A programmer is implementing the Singleton design pattern in a public class called Authenticator. What constructor(s) should the programmer declare in Authenticator?
a. Authenticator should have no explicitly declared constructors.
b. Authenticator should have a private constructor.
c. Authenticator should have a protected constructor.
d. Authenticator should have a public constructor, and a constructor with no access modifiers.

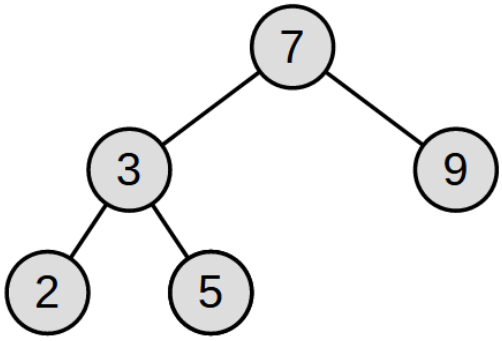
Q3[Software Testing]

Consider a program with the following control flow graph, where each oval represents a distinct statement.



What is the minimum number of test cases needed to achieve branch complete coverage for the shown control flow graph?
a. 4
b. 5
c. 6
d. 9
e. 12
f. 20
g. 21

Q4 [Data Structures]
Consider this AVL Tree, with keys sorted in ascending order, and root node 7.



When the key 6 is inserted, what type of rotation, if any, would be performed to balance the tree?
a. Left-right
b. Right-left
c. Left
d. Right
e. No rotation

Q5[Android]

Which of the following is the recommended way to support multiple languages in an Android app?

a. Detect the user's language using Java code and manually set text for each view.
b. Create separate layout files for each language and switch them in code.
c. Use string resources and provide translations in separate values folders (e.g., values-es/strings.xml).
d. Store all translated strings in a single strings.xml file and load them based on user input.

Q6[Tokeniser and Parser]

Consider the grammar with initial symbol ‘sentence’ and production rules given by

sentence → noun_phrase verb noun_phrase
noun_phrase → adjective noun_phrase | noun | noun_phrase “that” verb noun_phrase
noun → “cats” | “dogs” | “rabbits” | “people”
adjective → “white” | “black” | “gray” | “brown”

verb → “like” | “chase” | “talk to” | “see”
implemented with a tokeniser that ignores capitalisation and punctuaction, and a recursive descent parser.

When applied to the input “Dogs chase gray rabbits that talk to white rabbits.”, what node would be at the root of the parse tree?
a. a Sentence with verb=“chase”
b. a Sentence with verb=“talk to”
c. a Verb with string “chase”
d. a Verb with string “talk to”

Q7 [Persistent Data]

A client wishes to apply “sharding” to a large amount of data persistently. Sharding is the practice of splitting a single large file into a large number of smaller files, each of which can be accessed concurrently for greater performance. For example, instead of a file with 10,000 instances, the client wants to generate 10 files, each with 1,000 instances. Which formats are suitable for this purpose?
a. CSV is suitable, because it is non-hierarchical, so the parser does not need to know what tags are open at the start of a particular shard.
b. XML is suitable, because it supports attributes, which allow the developer to indicate where a shard ends and begins.
c. JSON is suitable, as it has native support for sharding through the GSON Java library.
d. Serialization is suitable only if the objects are each less than 128 bytes in size. In this case, Java can serialize multiple objects to the same file.

Q8[Refactoring]

```
class Vector2 {
     private double x, y;
     public Vector2(double x, double y) {
          this.x = x;
          this.y = y;
     }

     // Returns the distance from the origin to this vector.
     public double getLength() {
          return Math.sqrt(x*x + y*y);
     }

     // Scales the vectors together componentwise
     public Vector2 dilate(Vector2 other) {
          return new Vector2(x*other.x, y*other.y);
     }
}
```

Video game engines frequently use classes like the one partially shown above to represent a position in 2D space. A developer is tasked with extending this functionality by writing a Vector3 class to represent a position in 3D space by adding a z-axis.
What might the developer do in this situation to avoid violating the Liskov substitution principle?
a. Have Vector2 and Vector3 both extend an abstract Vector class.

b. Invert the class hierarchy by making Vector2 a subclass of Vector3.
c. Convert Vector2 from a class to an interface which is implemented by Vector3.
d. Make Vector2 and Vector3 both be immutable by marking their fields as final.

Q9 [Design by Contract]

```
/**
 * Interchanges the elements at indices i and j of the array.
 * (1)
 */
public static void <T> swap(T[] array, int i, int j) {
      T oldi = array[i];
      array[i] = array[j];
      array[j] = oldi;
}
```

The method swap interchanges the elements at indices i and j of an array. A developer wishes to write some JML annotations which can fill the label (1).

Which JML annotation has the correct syntax to give an informal description to the parameter i?
a. @parameter “The first index” i
b. @input i: “The first index.”
c. @param i The first index.
d. @argument The first index = i

Q10 [IP]

A colleague has found an implementation of an algorithm publicly available on GitHub, but there is no license listed accompanying the code. Your colleague wants to use this code in their work. Can they? What conditions must they adhere to?
a. The code is assumed to be under an exclusive license, and cannot legally be used.
b. The author is assumed to retain all copyright to the code, but because the code is publicly available, as long as appropriate attribution is provided, the code may legally be used.
c. The code is assumed to be under no license, and can legally be used without attribution. However, if the code contains a glitch or bug, the original author is not liable.
d. According to GitHub’s terms of service, public GitHub repositories without an explicit license are assumed to be under the GNU GPL v3 License. Your colleague may therefore use this code only if the modified source code is shared publicly.